

Fever, medically known as pyrexia or controlled hyperthermia, is a temporary increase in the body's core temperature, generally triggered by the immune system in response to an infection or inflammation. It is not a disease itself, but a symptom signaling that the body is fighting an underlying condition.

1. Medical Pathophysiology (The "Why")

Set Point Elevation: Fever occurs when the hypothalamus (the body's thermostat) raises its set point.

Pyrogens: Substances that induce fever are called pyrogens.

Exogenous Pyrogens: Microbes (bacteria, viruses) or their toxins enter the body.

Endogenous Pyrogens: The immune system (macrophages, monocytes) releases cytokines (e.g., Interleukin-1, Interleukin-6, TNF-alpha) in response to pathogens.

Prostaglandin E2 (

): Cytokines trigger the release of

in the hypothalamus, which acts on the thermoregulatory neurons, increasing the set point, resulting in chills and vasoconstriction to generate heat.

Purpose: Fever enhances immune cell activity, boosts interferon production, and inhibits the growth of bacteria and viruses that prefer normal body temperature.

2. Temperature Guidelines

Normal Body Temperature: Usually considered but normal ranges from

Fever: Typically defined as a temperature of

.

High Fever (Hyperpyrexia):

.

Low-Grade Fever:

3. Causes of Fever

Infections (Most Common): Viral fever (flu, COVID-19, dengue), bacterial infections (pneumonia, UTI), or fungal infections.

Inflammatory Conditions: Rheumatoid arthritis, lupus, or IBD (Crohn's/Colitis).

Immunization: Common reaction to vaccines (e.g., DTaP, COVID-19).

Non-Infectious/Environmental: Heatstroke, severe sunburn, blood clots, or drug-induced fever.

Malignancy: Leukemia, lymphoma.

4. Symptoms of Fever

High Temperature & Chills: Shivering (muscle contraction) occurs to raise heat.

Sweating/Flushed Skin: Occurs when the set point lowers.

Fatigue and Myalgia: Muscle aches and body weakness.

Headache.

Dehydration Symptoms: Thirst, dry mouth, reduced urine output.

Loss of Appetite.

Mental Status Changes: Confusion or hallucinations (in high fever).

5. Treatment and Management

Rest: Crucial for allowing the immune system to fight the infection.

Hydration: Water, clear broths, oral rehydration solutions to replace fluids lost via sweating and increased metabolism.

Tepid Sponging: Lukewarm water sponge bath can aid cooling; avoid cold water as it can cause shivering.

Remove Layers: Light clothing and bedding.

6. Medicine (Antipyretics)

Medication aims to lower the hypothalamic set point for comfort.

Paracetamol (Acetaminophen): Frequently used first-line for fever and pain.

NSAIDs (Ibuprofen/Naproxen): Effective for reducing fever and body aches.

Aspirin: Suitable for adults, but contraindicated in children due to Reye's syndrome risk.

Antibiotics: Only for bacterial infections.

7. Prevention

Hygiene: Frequent handwashing with soap.

Immunization: Staying up-to-date with vaccines.

Avoidance: Minimizing contact with infected individuals and taking environmental precautions like using insect repellent.

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Something went wrong with this response.

. Medical Definition and Pathology (The Disease)

Dengue is an acute arboviral (arthropod-borne) infection caused by the Dengue Virus (DENV), a positive-sense, single-stranded RNA virus from the Flaviviridae family.

Vector: Transmitted mainly by the bite of an infected female *Aedes aegypti* mosquito (rarely *Aedes albopictus*).

Serotypes: There are four distinct serotypes (DENV-1, DENV-2, DENV-3, DENV-4).

Pathogenesis:

Primary Infection: Infection with one serotype provides life-long immunity to that specific serotype but only temporary cross-protection against others.

Secondary Infection (Antibody-Dependent Enhancement - ADE): If a person is infected with a different serotype later, the antibodies from the first infection do not neutralize the new virus. Instead, they form a complex that helps the virus enter immune cells (monocytes/macrophages) more efficiently, leading to higher viral loads and severe disease.

Critical Mechanism: The virus causes high vascular permeability (leakage), leading to plasma loss (hemoconcentration), thrombocytopenia (low platelets), and coagulopathy (bleeding tendency).

Phases of Illness:

Febrile Phase (Days 1-3): Sudden high fever (up to 40°C), viremia.

Critical Phase (Days 3-7): Defervescence (fever breaks) but increased vascular leakage occurs. This is the danger zone.

Recovery Phase (Day 7+): Reabsorption of fluids.

2. Clinical Symptoms

Mild Dengue ("Break-bone fever"): High fever, severe headache, retro-orbital pain (pain behind the eyes), severe joint/muscle pain, skin rash (maculopapular), and minor bleeding (petechiae, gum bleeding).

Severe Dengue Warning Signs (Critical Phase):

Severe abdominal pain.

Persistent vomiting.

Mucosal bleeding (nasal, gingival, vaginal).

Rapid drop in temperature accompanied by rapid increase in hematocrit (hemoconcentration).

Lethargy or restlessness.

Liver enlargement.

Severe Dengue Complications: Dengue Shock Syndrome (DSS), Severe Hemorrhaging, Severe Organ Dysfunction (Hepatitis, Myocarditis, Encephalitis).

3. Diagnosis

NS1 Antigen Test: Best within the first 1-5 days of fever.

Serology (IgM/IgG): IgM ELISA tests are useful after day 5.

Complete Blood Count (CBC): Shows leukopenia (low white cells) followed by thrombocytopenia (platelets $<100,000/\text{mm}^3$) and high hematocrit.

4. Treatment and Medical Management

There is no specific antiviral medicine for dengue; treatment is entirely supportive.

Hospitalization Criteria: Patients with warning signs, dehydration, or comorbidities (pregnancy, elderly).

Fluid Management:

Mild Cases: Oral Rehydration Solution (ORS), fruit juices, water, coconut water to maintain hydration.

Severe Cases: IV fluids (Crystalloids like Ringer's lactate or Normal Saline) are the cornerstone of management. Over-hydration must be avoided in the recovery phase to prevent pulmonary edema.

Monitoring: Frequent hematocrit and platelet count monitoring during the critical

phase.

Blood Transfusion: Used only in cases of severe hemorrhage.

5. Medicine and Pharmacology

Safe Medications:

Acetaminophen/Paracetamol: For fever and pain management.

Medications to AVOID:

NSAIDs (Ibuprofen, Naproxen, Diclofenac): Avoid completely.

Aspirin/Salicylates: Avoid completely.

Reason: NSAIDs/Aspirin increase the risk of bleeding (due to thrombocytopenia) and gastric injury.

Vaccines:

Dengvaxia (CYD-TDV): Licensed for 9-16 year olds with a prior laboratory-confirmed infection.

Qdenga (TAK-003): Second-gen, approved in several countries for ages 4-60, regardless of serostatus.

6. Prevention (Vector Control)

Personal Protection: Use EPA-registered insect repellents (containing 20%-30% DEET, Picaridin, or IR3535). Wear long-sleeved shirts/pants.

Environmental Control: Eliminate standing water in pots, tires, and gutters (Aedes mosquitoes breed in clean, stagnant water).

Household Barriers: Use window screens and air conditioning.

Indoor Protection: Use insecticide-treated mosquito nets, especially when sleeping during the day (as Aedes mosquitoes are daytime biters).

Community: Support fogging and larvicide application in endemic areas.

Medical Notes: Typhoid Fever (Enteric Fever)

1. Disease Definition and Etiology

Definition: Typhoid fever is a life-threatening systemic infection caused by the bacterium *Salmonella enterica* serotype Typhi (*S. Typhi*). It is commonly referred to as "enteric fever" because it primarily infects the intestinal tract, though it spreads throughout the body via the bloodstream.

Etiological Agent: *Salmonella Typhi* (and *Salmonella Paratyphi*, which causes a milder form of the disease).

Transmission: Fecal-oral route. The bacteria are shed in the feces (and occasionally urine) of infected people or chronic carriers. It is contracted by ingesting contaminated food or water.

Carrier State: Roughly 3-5% of infected individuals become long-term carriers, storing the bacteria in their gallbladder and shedding them for over a year, even after symptoms subside.

2. Pathophysiology (How the disease works)

Ingestion: The bacteria are ingested and survive the gastric acid in the stomach.

Invasion: They move to the small intestine and penetrate the mucosa.

Immune Evasion: Inside the intestinal wall, they are engulfed by macrophages (immune cells) but survive and multiply within them.

Systemic Spread: The infected macrophages travel through the lymphatic system to the bloodstream (bacteremia), spreading the bacteria to the liver, spleen, and bone

marrow.

Re-invasion: Bacteria are released back into the intestine, attacking Peyer's patches (lymphatic tissue in the intestines), leading to inflammation, necrosis, and potentially perforation.

3. Symptoms and Clinical Stages

Symptoms usually appear 1-3 weeks after exposure.

Stage 1 (Week 1): Insidious onset, gradually increasing "step-ladder" fever (rising every day), persistent headache, dry cough, malaise.

Stage 2 (Week 2): High fever (103°F-104°F) persists. "Rose spots" (a faint pink rash on the chest/abdomen) may appear. Significant abdominal pain, splenomegaly (enlarged spleen), and hepatomegaly (enlarged liver). Constipation is more common in adults, while diarrhea is more common in children.

Stage 3 (Week 3 - Severe): Untreated patients may become delirious (the "typhoid state"). Complications are highly likely here: intestinal perforation, internal bleeding (hemorrhage), and sepsis.

Stage 4 (Week 4): Symptoms begin to subside in survivors, but extreme weakness persists.

4. Diagnosis

Blood Culture: Gold standard, particularly in the first week, although sensitivity varies (40-60%).

Stool/Urine Culture: Confirms presence of bacteria, especially in carriers.

Bone Marrow Culture: Most accurate test, but rarely done due to pain.

Widal Test: Measures antibodies. Used in endemic areas, but has low specificity and many false positives.

Typhidot/Tubex: Rapid diagnostic tests to detect IgM/IgG antibodies.

5. Treatment and Medicine

Typhoid must be treated with antibiotics. Self-medication is discouraged due to high rates of antimicrobial resistance (MDR and XDR strains).

First-line Treatment (Uncomplicated):

Azithromycin: Often preferred (1g on day 1, then 500mg for 7 days).

Cefixime: Oral 10-14 day course.

Severe Cases (Hospitalization):

Ceftriaxone: IV parenteral therapy for 5-7 days.

Extensively Drug-Resistant (XDR) Typhoid:

Meropenem: Or other carbapenems.

Adjunctive Therapy: Steroids (e.g., Dexamethasone) may be used for severe toxemia or delirium.

Supportive Care: Fluid replacement (ORS), antipyretics (Paracetamol) for fever.

6. Prevention

Vaccination:

Typhoid Conjugate Vaccine (TCV): Recommended by WHO for children 6 months+ and adults up to 65. Single shot, long-lasting immunity.

Vi Polysaccharide Vaccine (ViCPS): Injectable, for age 2+, booster every 2 years.

Live Oral Vaccine (Ty21a): For age 6+, 4 capsules (alternate days), booster every 5

years.

Hygiene and Food Safety:

"Boil it, cook it, peel it, or forget it": Drink bottled or boiled water.

Hand Hygiene: Frequent handwashing with soap, especially before eating and after using the bathroom.

Avoid: Raw vegetables, street food, ice, and unpasteurized dairy.

7. Complications (Untreated)

Intestinal Perforation/Hemorrhage: Occurs in 3rd week, requiring emergency surgery.

Typhoid Encephalopathy: Cerebral dysfunction.

Sepsis: Bloodstream infection.

Myocarditis: Heart muscle inflammation.

Malaria: Medical Notes

1. Definition and Etiology

Malaria is an acute, potentially life-threatening infectious disease caused by protozoan parasites of the genus *Plasmodium*. It is transmitted to humans through the bite of an infected female *Anopheles* mosquito.

Causative Agents (Parasite Species):

Plasmodium falciparum: Most dangerous, causes the majority of severe disease and death, dominant in Africa.

Plasmodium vivax: Most geographically widespread, causes relapsing malaria due to dormant liver forms (hypnozoites).

Plasmodium ovale: Causes relapsing malaria; dormant liver stages.

Plasmodium malariae: Causes chronic, low-grade infections.

Plasmodium knowlesi: Zoonotic species (macaques to humans), causes severe malaria in Southeast Asia.

2. Pathophysiology (Disease Mechanism)

The pathogenesis involves a complex lifecycle alternating between the mosquito and human host.

A. Life Cycle in Humans

Inoculation (Sporozoite Stage): Infected mosquito injects sporozoites into the bloodstream.

Exo-erythrocytic (Liver Stage): Sporozoites travel to the liver, invade hepatocytes, and reproduce asexually, turning into merozoites. *P. vivax* and *P. ovale* can form dormant hypnozoites here.

Erythrocytic (Blood Stage): Merozoites are released, invade red blood cells (RBCs), mature into trophozoites, and then schizonts, which rupture the RBCs, releasing more merozoites.

Clinical Disease: The rupture of RBCs is the cause of periodic fever, hemolysis (destruction of RBCs), and anemia.

B. Severe Malaria Pathogenesis

Cytoadherence and Sequestration: *P. falciparum* infected RBCs express adhesion proteins (pFEMP1), causing them to stick to vascular endothelium (small blood vessels).

Tissue Hypoxia: Obstructed blood flow (microvascular obstruction) leads to organ hypoxia and dysfunction, notably cerebral malaria.

Inflammatory Response: Release of parasite toxins stimulates high levels of

pro-inflammatory cytokines (TNF-
), causing severe fever and systemic damage.

3. Symptoms and Clinical Presentation

Symptoms typically appear 10-15 days after the bite, but can be delayed weeks to months.

A. Uncomplicated Malaria (Initial Presentation)

Classic Malaria Paroxysm: A cyclical sequence of:

Cold Stage: Shivering, chills.

Hot Stage: High fever (up to 41°C), headache, vomiting.

Sweating Stage: Rapid drop in temperature, profuse sweating.

Non-specific Symptoms: Fatigue, malaise, myalgia (muscle aches), arthralgia, abdominal pain, diarrhea.

B. Severe Malaria (Medical Emergency)

Cerebral Malaria: Altered consciousness, coma, seizures.

Severe Anemia: Hemoglobin < 7 g/dL due to extensive RBC destruction.

Acute Kidney Injury (AKI): Oliguria/anuria.

Metabolic Acidosis/Respiratory Distress: Rapid or deep breathing (ARDS).

Hypoglycemia: Low blood sugar, often in pregnant women and children.

Blackwater Fever: Severe intravascular hemolysis, causing dark-colored urine.

4. Diagnosis and Investigations

Prompt diagnosis is crucial to prevent progression to severe malaria.

Microscopy (Gold Standard): Thick and thin blood smears stained with Giemsa stain to identify the species and calculate parasitemia (percentage of RBCs infected).

Rapid Diagnostic Tests (RDTs): Detect specific malaria antigens (HRP2 or LDH). They are rapid and useful in areas without microscopy.

Other Lab Findings: Thrombocytopenia (low platelets), elevated LDH, elevated bilirubin, jaundice.

5. Treatment and Medicines

Treatment is based on species, severity, drug resistance, and patient weight.

World Health Organization (WHO)

World Health Organization (WHO)

A. Uncomplicated *P. falciparum* Malaria

Artemisinin-based Combination Therapy (ACTs): ACT is the recommended first-line treatment.

Examples: Artemether + Lumefantrine (Coartem), Artesunate + Amodiaquine, Dihydroartemisinin + Piperaquine.

B. Uncomplicated *P. vivax*/*P. ovale* Malaria

Chloroquine (CQ): First-line in areas where CQ is still effective.

ACT: Recommended in CQ-resistant areas.

Radical Cure (Relapse Prevention): Primaquine or Tafenoquine must be added to treat dormant liver forms (hypnozoites).

C. Severe Malaria

First-line: Intravenous (IV) or intramuscular (IM) Artesunate for at least 24 hours.

Alternative: IV Quinine or IM Artemether if artesunate is unavailable.

Follow-on: Once stable, follow with a 3-day course of oral ACT.

⚠ Crucial Medical Warning: Primaquine is contraindicated in patients with Glucose-6-phosphate dehydrogenase (G6PD) deficiency, as it can cause fatal hemolysis.

6. Prevention

Vector Control:

Insecticide-treated nets (ITNs/LLINs): Key protection method, particularly for children and pregnant women.

Indoor Residual Spraying (IRS): Spraying insecticide on walls.

Chemoprophylaxis (for travelers):

Atovaquone-proguanil (Malarone), Doxycycline, or Mefloquine are taken before, during, and after travel.

Vaccines: RTS,S/AS01 and R21/Matrix-M vaccines are recommended for children in high-transmission areas.

Personal Protection: Using repellent (DEET, Icaridin), covering skin after dusk.

Diabetes Mellitus: Detailed Medical Notes

1. Definition and Disease Overview

Diabetes Mellitus (DM) is a chronic, metabolic disorder characterized by persistent hyperglycemia (high blood sugar). It results from deficiencies in insulin secretion, reduced insulin action (insulin resistance), or both.

Insulin's Role: Produced by

-cells in the pancreas, insulin allows glucose to enter cells for energy.

Pathophysiology: When insulin is absent or ineffective, glucose builds up in the blood, causing hyperglycemia, while body cells become energy-deprived.

Long-Term Complications: Chronic hyperglycemia damages blood vessels (macrovascular and microvascular) and nerves.

2. Classification

Type 1 Diabetes (T1DM): An autoimmune disease where the immune system destroys -cells, leading to absolute insulin deficiency. Usually diagnosed in children/young adults, but can occur at any age.

Type 2 Diabetes (T2DM): The most common form (over 90% of cases). Characterized by a combination of insulin resistance (cells don't respond to insulin) and relative insulin deficiency. Associated with obesity, sedentary lifestyle, and genetics.

Gestational Diabetes (GDM): High blood glucose that develops during pregnancy, typically resolving post-delivery, but increases the risk of developing T2DM later.

Prediabetes: A condition where blood sugar levels are higher than normal but not high enough for a T2DM diagnosis.

3. Symptoms and Clinical Presentation

Symptoms arise from high glucose levels (hyperglycemia) and lack of cellular energy.

Classic Triad (More common in T1DM):

Polyuria: Frequent urination (due to osmotic diuresis).

Polydipsia: Excessive thirst (due to dehydration).

Polyphagia: Excessive hunger (due to cellular starvation).

Other Symptoms:

Unexplained weight loss (common in T1DM).

Fatigue and weakness.

Blurred vision (due to lens changes from sugar).

Slow-healing sores or infections (skin, gum, vaginal yeast).

Numbness or tingling in hands/feet (neuropathy).

Darkened skin patches (acanthosis nigricans - associated with insulin resistance).

4. Diagnosis and Monitoring

HbA1c (Glycated Hemoglobin): Measures average blood sugar over 2-3 months.

6.5% indicates diabetes.

Fasting Plasma Glucose (FPG):

126 mg/dL after 8-12 hours of fasting.

Oral Glucose Tolerance Test (OGTT): 2-hour blood glucose

200 mg/dL after a 75-g glucose drink.

Random Blood Glucose:

200 mg/dL + symptoms of hyperglycemia.

5. Treatment and Management

The primary goal is to maintain glycemic control to prevent complications.

A. Lifestyle Management (Cornerstone of Therapy)

Diet: Balanced, low-GI (Glycemic Index), high-fiber diet focusing on vegetables, whole grains, and lean protein.

Exercise: At least 150 minutes of moderate-intensity aerobic activity per week, plus resistance training.

Weight Loss: Losing 5-7% of body weight significantly improves T2DM management.

B. Pharmacological Treatment (Medicines)

Type 1: Requires exogenous insulin therapy (lifelong).

Type 2: Stepwise approach, often starting with oral agents.

Biguanides (Metformin): First-line therapy for T2DM. Decreases hepatic glucose production and improves insulin sensitivity.

GLP-1 Receptor Agonists (e.g., Semaglutide, Liraglutide): Enhance insulin secretion, reduce glucagon, and aid in weight loss. Recommended for those with ASCVD (heart disease) or MASLD (liver disease).

SGLT2 Inhibitors (e.g., Empagliflozin, Dapagliflozin): Promote glucose excretion in urine. Beneficial for heart failure and chronic kidney disease.

DPP-4 Inhibitors (e.g., Sitagliptin): Increase incretin levels to stimulate insulin release.

Sulfonylureas (e.g., Glimepiride): Stimulate the pancreas to secrete more insulin.

Insulin Therapy: Used in T2DM when oral agents are insufficient or in advanced disease.

C. Advanced Management

Continuous Glucose Monitoring (CGM): Recommended for all insulin-treated patients at diagnosis to monitor trends.

Automated Insulin Delivery (AID) Systems: Technology that automates insulin delivery based on CGM readings.

6. Complications

Acute: Diabetic Ketoacidosis (DKA - mostly T1D) and Hyperosmolar Hyperglycemic State (HHS - mostly T2D). Both are medical emergencies.

Chronic (Macrovascular): Heart attack, stroke, peripheral vascular disease.

Chronic (Microvascular): Retinopathy (blindness), Nephropathy (kidney failure), Neuropathy (nerve damage).

7. Prevention

Type 1: Cannot currently be prevented.

Type 2: Can be prevented or delayed through weight management, healthy diet, and regular exercise.

Proactive Steps: Smoking cessation, moderate alcohol intake, and regular screening for high-risk individuals.

High Blood Pressure: Hypertension

Hypertension is a chronic disease often called the "silent killer" because it typically causes no symptoms until significant vascular damage has occurred.

Disease Mechanisms: Long-term high pressure creates wear and tear on artery walls, leading to atherosclerosis (hardening), tiny cracks, and reduced elasticity, increasing the workload on the heart.

Symptoms (When severely high): Headaches, nosebleeds, dizziness, shortness of breath, chest pain, vision changes.

Causes: High sodium intake, sedentary lifestyle, obesity, genetics, smoking, and stress.

Treatment & Medicine:

Lifestyle: DASH diet, reduced sodium (under 1,500-2,300mg daily), regular aerobic exercise, smoking cessation, alcohol limitation.

Medication: Diuretics ("water pills"), ACE inhibitors (e.g., Lisinopril), ARBs (e.g., Losartan), Calcium channel blockers (e.g., Amlodipine), Beta-blockers.

Prevention: Maintain a healthy weight, consistent exercise (30+ mins/day), and limiting processed foods.

Low Blood Pressure: Hypotension

Hypotension occurs when the heart, brain, and other vital organs do not receive enough blood flow.

Disease Mechanism: Reduced peripheral vascular resistance or low blood volume prevents adequate tissue perfusion.

Symptoms: Dizziness, lightheadedness, fainting (syncope), blurred vision, fatigue, nausea.

Causes: Dehydration, severe bleeding (shock), heart problems (very slow/fast heart rate), endocrine problems, or medication side effects.

Treatment: Increasing fluid intake, increasing salt intake (under medical advice), wearing compression stockings, and medication to increase blood volume or tighten

blood vessels.

Prevention: Hydration, rising slowly from sitting/lying down, smaller, more frequent meals.

Summary of Differences

Target Reading: Normal BP is mmHg.

Hypertension Danger: Organ damage due to excessive pressure.

Hypotension Danger: Organ dysfunction due to inadequate oxygen delivery.

Management: Both require consistent monitoring and medical

1. Medical Definition and Overview

What is Loose Motion? Medically termed diarrhea, it is defined as the passage of three or more loose, watery, or unformed stools within a 24-hour period.

Mechanism: It occurs when the intestines do not absorb enough water, or secrete too much fluid, often due to irritation or inflammation, forcing waste through the digestive tract too quickly.

Types:

Acute Diarrhea: Lasts 1-2 days, usually self-limiting (viral/food poisoning).

Persistent Diarrhea: Lasts 2-4 weeks.

Chronic Diarrhea: Lasts more than 4 weeks; indicates an underlying disease.

2. Etiology (Causes)

Loose motion is generally a symptom of another condition rather than a disease itself.

Infections (Most Common):

Viral Gastroenteritis: Norovirus (adults) and Rotavirus (children).

Bacterial Infections: E. coli, Salmonella, Campylobacter, Shigella (often via contaminated food/water—"food poisoning").

Parasitic Infections: Giardia lamblia or Entamoeba histolytica.

Medication Side Effects: Antibiotics (disrupt gut flora), magnesium-containing antacids, and chemotherapy drugs.

Dietary Factors & Intolerances: Lactose intolerance (dairy), artificial sweeteners (sorbitol), or food allergies.

Chronic Diseases: Irritable Bowel Syndrome (IBS), Inflammatory Bowel Disease (IBD: Crohn's disease, ulcerative colitis), celiac disease.

Stress & Anxiety: High emotional stress can disrupt bowel motility.

3. Symptoms and Diagnosis

Common symptoms include:

Frequent loose or watery stools.

Abdominal cramps and pain.

Nausea and vomiting.

Urgent need to use the bathroom.

Bloating and rumbling in the tummy.

Fever and chills (suggests infection).

Presence of mucus or blood in the stool (dysentery).